$$2(Ax-b)^{\mathrm{T}}A = 0.$$

Recalling exercise 18.25(2), we rewrite this as

$$A^{\mathrm{T}}(Ax-b) = 0.$$

i.e.

$$A^{\mathrm{T}}Ax = A^{\mathrm{T}}b.$$

If $A^{\mathrm{T}}A$ is non-singular, it follows that there is only one candidate for a minimising x, namely

$$x = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b.$$

In the preceding example, the geometry of the situation makes it clear that a minimising x must exist. However, a geometrical argument is no substitute for a proof and, in any case, the geometry of the situation is seldom as transparent as in example 19.30. Some method is therefore necessary for classifying stationary points into local maxima, local minima, saddle points or other more exotic categories. Our experience with the real case would indicate an examination of the *second* derivative at a stationary point. But how is this defined?

19.31 Second derivatives

We consider only the second derivative of *real-valued* functions. The reason for this is that the derivative of a real-valued function is a vector and we know how to differentiate vector-valued functions. In order to discuss second derivatives in general, we should need first to discuss the derivative of matrix-valued functions and for this purpose the notation of matrix algebra is not adequate. (See §19.48.)

Even restricting ourselves to real-valued functions, we have a small problem since §19.14 assumes that we have a column vector to differentiate whereas $f'(x)$ is a row vector. However, this problem is easily overcome. If $\phi(x)$ is a row vector we define $\phi'(x)$ by

$$\phi'(x)^{\mathrm{T}} = \frac{d}{dx}\{\phi(x)^{\mathrm{T}}\}$$

provided the latter derivative exists. If ϕ is differentiable at ξ, we then have that

$$\phi(x)-\phi(\xi)-(x-\xi)^{\mathrm{T}}\phi'(\xi) = o(x-\xi)$$

i.e. the theory for row vectors is the same as that for column vectors except that it is necessary to write $(x-\xi)^{\mathrm{T}}\phi'(\xi)$ rather than $f'(\xi)(x-\xi)$ wherever the latter expression occurs.

With this understanding, we can now define the second derivative of a real-valued function f at the point x by

$$f''(x)^{\mathrm{T}} = \frac{d}{dx} f'(x)^{\mathrm{T}}$$

provided the latter derivative exists. Since

$$f'(x)^{\mathrm{T}} = \begin{pmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_n(x) \end{pmatrix}$$

it follows immediately from theorem 19.16 that $f''(x)$ is the $n \times n$ matrix

$$f''(x) = \begin{pmatrix} f_{11}(x) & f_{12}(x) \ldots f_{1n}(x) \\ f_{21}(x) & f_{22}(x) \ldots f_{2n}(x) \\ \ldots\ldots\ldots\ldots\ldots\ldots \\ f_{n1}(x) & f_{n2}(x) \ldots f_{nn}(x) \end{pmatrix} \tag{1}$$

where

$$f_{ij}(x) = \frac{\partial}{\partial x_i}\{f_j(x)\} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} f(x)\}.$$

It is customary to abbreviate the latter notation and to write

$$f_{ij}(x) = \frac{\partial^2}{\partial x_i \partial x_j} f(x).$$

Thus, if $y = f(x)$,

$$\frac{d^2y}{dx^2} = \begin{pmatrix} \dfrac{\partial^2 y}{\partial x_1^2} & \dfrac{\partial^2 y}{\partial x_1 \partial x_2} \cdots \dfrac{\partial^2 y}{\partial x_1 \partial x_n} \\ \dfrac{\partial^2 y}{\partial x_2 \partial x_1} & \dfrac{\partial^2 y}{\partial x_2^2} \quad \cdots \dfrac{\partial^2 y}{\partial x_2 \partial x_n} \\ \vdots \\ \dfrac{\partial^2 y}{\partial x_n \partial x_1} & \dfrac{\partial^2 y}{\partial x_n \partial x_2} \cdots \dfrac{\partial^2 y}{\partial x_n^2} \end{pmatrix}$$

and so the second derivative of a real-valued function consists of an $n \times n$ matrix whose entries consist of all second order partial derivatives of the function. From theorem 19.17 we know that a sufficient condition that f be twice differentiable at a point ξ is that all these second order partial derivatives exist and are *continuous* at ξ.

19.32 Example Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1$. We have that

$$\frac{\partial y}{\partial x_1} = 2x_1x_2 + x_2^3 \qquad \frac{\partial y}{\partial x_2} = x_1^2 + 3x_2^2x_1$$

and so

$$\frac{\partial^2 y}{\partial x_1^2} = 2x_2 \qquad \frac{\partial^2 y}{\partial x_1 \partial x_2} = 2x_1 + 3x_2^2$$

$$\frac{\partial^2 y}{\partial x_2 \partial x_1} = 2x_1 + 3x_2^2 \qquad \frac{\partial^2 y}{\partial x_2^2} = 6x_1x_2.$$

All these are continuous everywhere and hence f is twice differentiable at every point with

$$f''(x_1, x_2) = \begin{pmatrix} 2x_2 & 2x_1 + 3x_2^2 \\ 2x_1 + 3x_2^2 & 6x_1x_2 \end{pmatrix}.$$

It is no accident that the matrix $f''(x_1, x_2)$ in the preceding example is *symmetric*, i.e.

$$\frac{\partial^2 y}{\partial x_1 \partial x_2} = \frac{\partial^2 y}{\partial x_2 \partial x_1}.$$

19.33 Proposition If all the second order partial derivatives in the matrix (1) of §19.31 are continuous at ξ, then the $n \times n$ matrix $f''(\xi)$ is symmetric, i.e. for all i and j

$$f_{ij}(\xi) = f_{ji}(\xi).$$

19.34 Mean value theorems

The mean value theorem of §11.5 generalises immediately to the case of *real-valued* functions of a vector variable. Note, however, that it does *not* generalise to vector-valued functions. (See exercise 19.47(5).)

19.35 Theorem (mean value theorem) Let L denote the straight line segment joining $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^n$ and let L^0 be the set obtained by deleting a and b from L. If f is a real-valued function which is continuous at each point of L and differentiable at each point of L^0, then

$$f(b) - f(a) = f'(\xi)(b - a)$$

for some $\xi \in L^0$.

Proof Define F: $[0, 1] \to \mathbb{R}$ by

$$F(t) = f((1 - t)a + tb) \; \mathbb{R} \; (0 \leqslant t \leqslant 1).$$

Then F is continuous on $[0, 1]$ and differentiable on $(0, 1)$. (See exercise 18.20(5).) By theorem 11.6

$$\frac{F(1)-F(0)}{1-0} = F'(\tau)$$

for some $\tau \in (0, 1)$. Thus $f(b)-f(a) = F(1)-F(0) = F'(\tau)$. It remains to express $F'(\tau)$ in terms of the function f. Write $y = f(x)$ and $x = (1-t)a + tb$. Then

$$F'(t) = \frac{dy}{dt} = \frac{dy}{dx}\cdot\frac{dx}{dt} = f'(x)(b-a).$$

Writing $\xi = (1-\tau)a + \tau b$, it follows that

$$f(b)-f(a) = F'(\tau) = f'(\xi)(b-a).$$

In chapter 11, we went on to prove Taylor's theorem which, as we pointed out, may be regarded as the nth order mean value theorem. This also generalises to the case of *real-valued* functions but we consider only the case $n = 2$ since we have not discussed derivatives of higher order.

19.36 Theorem (Taylor's theorem of order 2) Suppose that f is a real-valued function which is twice differentiable at each point $x \in \mathbb{R}^n$ which satisfies $\|x-\xi\| < \Delta$. Then for any such x,

$$f(x) = f(\xi) + \frac{1}{1!}f'(\xi)(x-\xi) + \frac{1}{2!}(x-\xi)^{\mathrm{T}}f''(\eta)(x-\xi)$$

for some η on the line segment L^0 between x and ξ.

Proof We apply theorem 11.10 to the function F: $[0, 1] \to \mathbb{R}$ defined by $F(t) = f((1-t)\xi + tx)$ $(0 \leqslant t \leqslant 1)$ and obtain

$$F(1) = F(0) + \frac{1}{1!}F'(0) + \frac{1}{2!}F''(\tau)$$

for some τ satisfying $0 < \tau < 1$. We have that $F(1) = f(x)$, $F(0) = f(\xi)$ and, as in the previous theorem,

$$F'(t) = f'((1-t)\xi + tx)(x-\xi)$$

so that $F'(0) = f'(\xi)(x-\xi)$. It remains to express $F''(\tau)$ in terms of f. This requires a little care with transposition signs. (See exercise 18.25(2).) Observe that

$$F''(t)^{\mathrm{T}} = \frac{d}{dt}\{F'(t)\}^{\mathrm{T}}$$

$$= \frac{d}{dt}\{f'((1-t)\xi + tx)(x-\xi)\}^{\mathrm{T}}$$